DO NOW — RECALL

Introduction to time and space complexity

Name: _____ Class: _____ Date: _____

1. What makes a binary search trace convincing?

2. State the average-case time complexity of quick sort for n items.

3. How many values are in an int array of length n?

4. Apply an eight-bit left shift by two to 00010110_2 and bitwise AND to 00110110_2 with 00001111_2. Give both results in binary.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Showing low, high, mid, comparisons and any bound updates.

2. Model answer

O(n log n).

3. Model answer

n values; storing it needs O(n) space.

4. Model answer

Left shift 00010110 << 2 = 01011000 (eight-bit, zero-filled). AND 00110110 ∧ 00001111 = 00000110.

Vocabulary — know these terms

time complexity — Asymptotic growth in algorithm work.

auxiliary space — Extra memory used by an algorithm.

Big O — Upper asymptotic bound commonly used to classify growth.

quadratic — Growth proportional to n^2.

exponential — Growth such as 2^n.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain the answer

Quadratic count rises from 64 to 256, multiplying by four; this difference grows with n.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Analyse the loops

One full loop runs n iterations; a pair of n-sized nested loops performs n^2 inner iterations.

Separate time and space

Big O time counts growth of work; auxiliary space measures extra storage beyond the original input.

Compare feasible scales

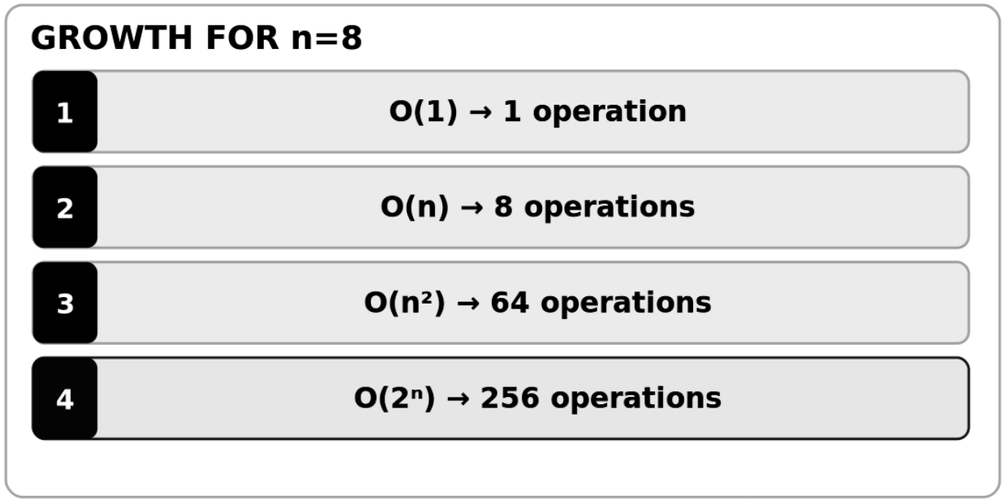
At n=16, exponential growth can be dramatically more expensive than linear growth. Big O omits constants and measured run time.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

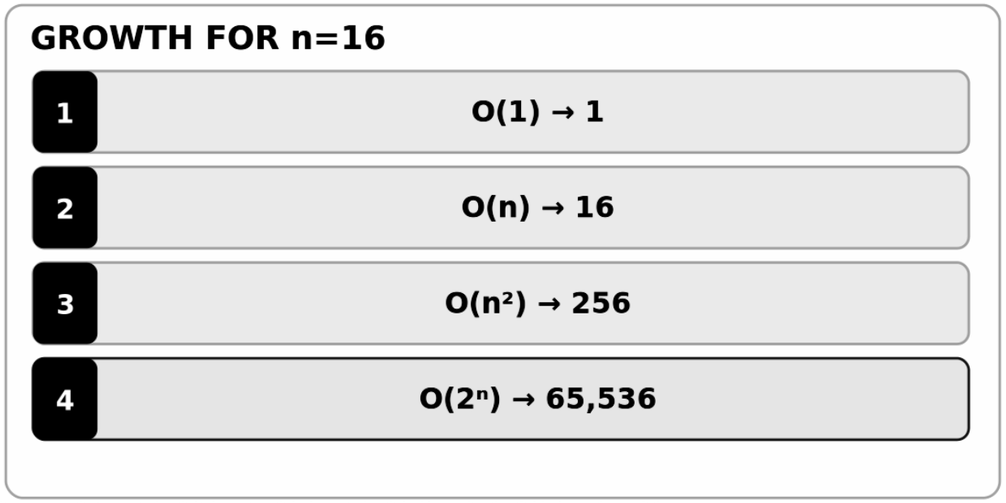
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



GROWTH FOR n=8; O(1) → 1 operation; O(n) → 8 operations; O(n²) → 64 operations; O(2ⁿ) → 256 operations

New Learning



GROWTH FOR n=16; O(1) → 1; O(n) → 16; O(n²) → 256; O(2ⁿ) → 65,536

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

O(n) time necessarily means an algorithm uses O(n) additional memory.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

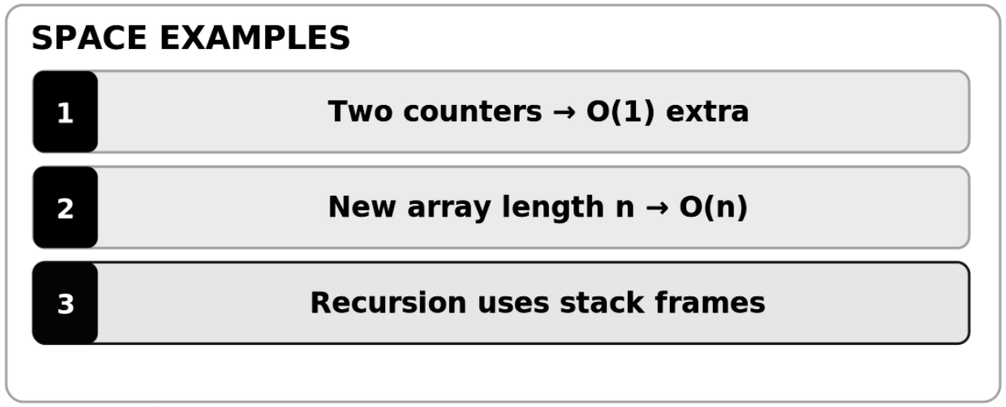
Question 3

A method uses an in-place O(n²) sort and another uses merge sort with O(n log n) time and O(n) auxiliary space. Explain one situation favouring each.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



SPACE EXAMPLES; Two counters → O(1) extra; New array length n → O(n); Recursion uses stack frames

I DO — WORKED EXAMPLE

Follow the worked method

For n=8, compare one loop visiting each item with two full nested loops.

Predict first

One scan has eight iterations, giving linear growth O(n).

Explain the answer

Nested loops perform 8×8=64 inner iterations, giving quadratic growth O(n²).

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State the Big O of (a) constant assignment, (b) one n-element loop, (c) two full nested n-loops and (d) repeatedly halving an interval.

Question 2

A program copies an n-element array into a new n-element array. Give its time and auxiliary-space complexity and justify both.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The linear count doubles from 8 to 16.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

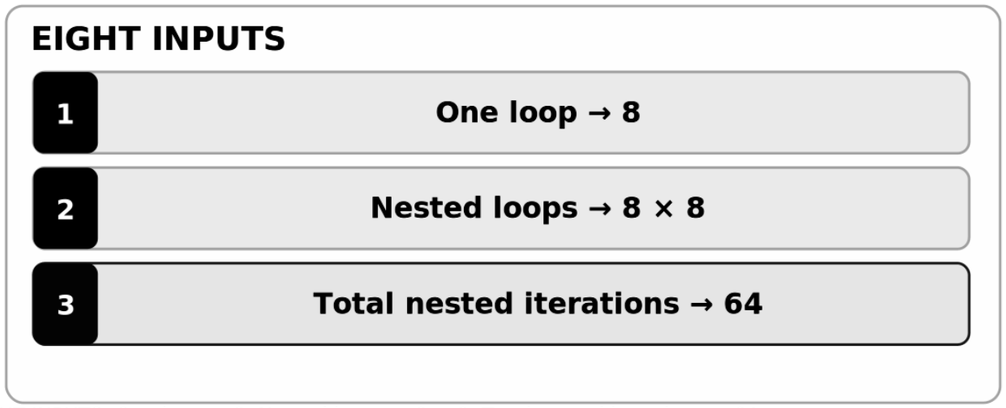
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For n=8, compare one loop visiting each item with two full nested loops.

One scan has eight iterations, giving linear growth O(n).



EIGHT INPUTS; One loop → 8; Nested loops → 8 × 8; Total nested iterations → 64

WE DO — GUIDED PRACTICE

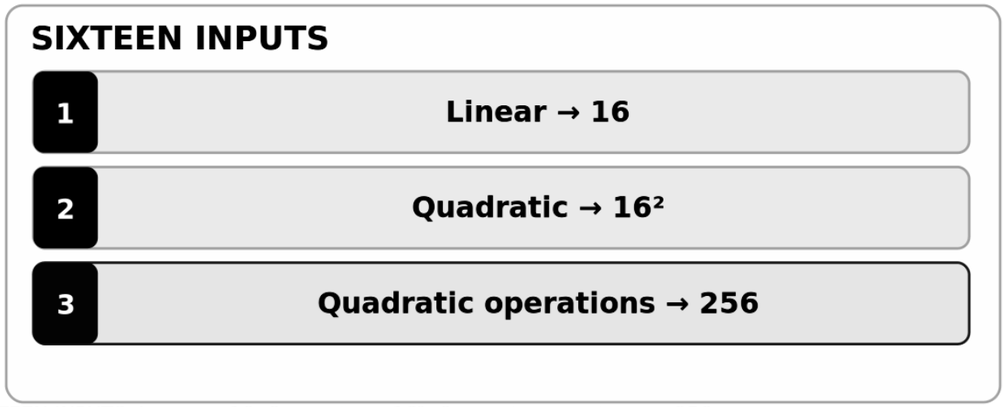
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For n=16, compare linear and quadratic operation counts. What happens when n doubles from 8?

The linear count doubles from 8 to 16.



SIXTEEN INPUTS; Linear → 16; Quadratic → 16²; Quadratic operations → 256